

Linear Search

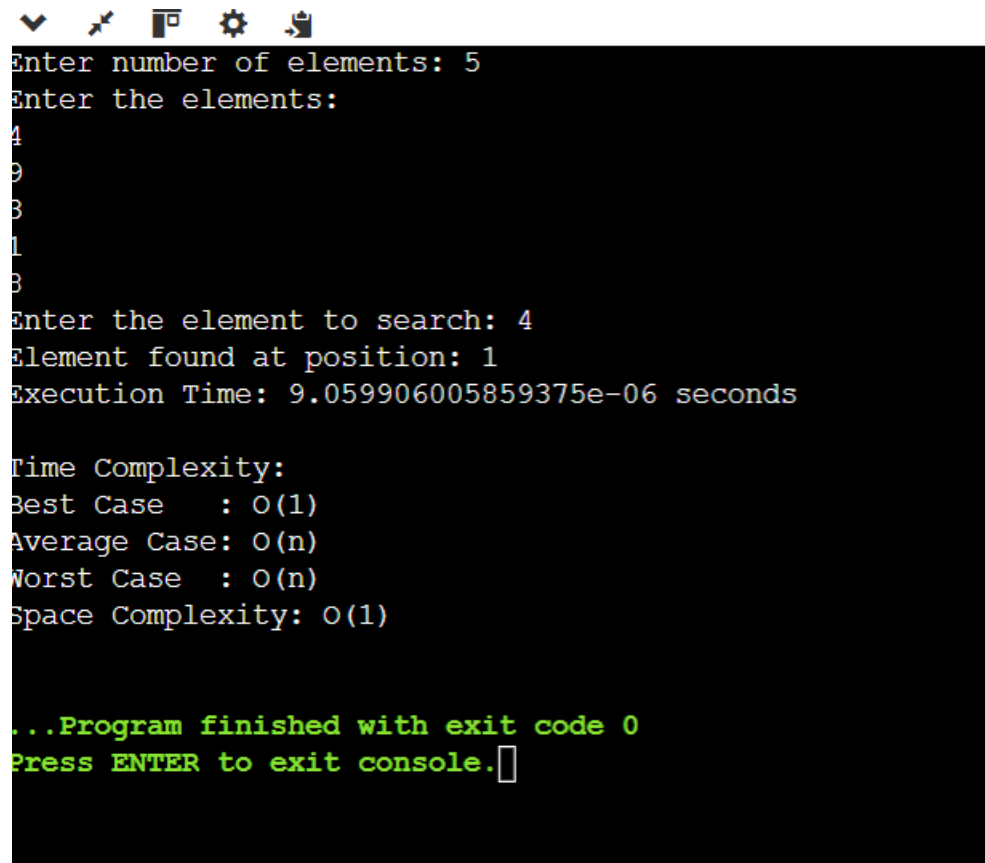
Definition

Linear Search is a searching algorithm that checks each element of an array **one by one from the beginning** until the required element is found or the array ends.

Algorithm

1. Start from the first element.
2. Compare the current element with the **target value**.
3. If they are equal, return the position/index.
4. If not equal, move to the next element.
5. Repeat until the element is found or the array ends.
6. If the element is not found, return **-1**.

Output:

A terminal window with a black background and white text. At the top, there are five small icons: a downward arrow, a cursor, a document, a gear, and a trash can. The text in the terminal shows the program's execution flow: entering 5 elements, listing them (4, 9, 3, 1, 8), searching for 4, finding it at position 1, showing execution time, and listing time and space complexities. It ends with a green prompt to press ENTER to exit.

```
Enter number of elements: 5
Enter the elements:
4
9
3
1
8
Enter the element to search: 4
Element found at position: 1
Execution Time: 9.059906005859375e-06 seconds

Time Complexity:
Best Case    : O(1)
Average Case: O(n)
Worst Case   : O(n)
Space Complexity: O(1)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

- **Best Case:** $O(1)$ — element is found at the first position.
- **Average Case:** $O(n)$ — element is somewhere in the middle.
- **Worst Case:** $O(n)$ — element is at the last position or not present.

Space Complexity

- **Auxiliary Space:** $O(1)$
- No extra array is required.